AI & Agents — 1-Week Intensive Syllabus

> LLMs + Agents: Beginner → Senior in 7 Days

🎯 Weekly Goal

By the end of this week, you will build a fully working Agent system with planning, tools, memory (RAG), validation loops, and multi-agent orchestration — the foundation of a Senior-level understanding.

Prerequisites

- Python 3.10+ installed
- Basic understanding of machine learning concepts (what a neural network is)
- Comfort with APIs (REST, JSON)
- `pip install openai tiktoken numpy chromadb` (installed as needed per day)

Daily Schedule

Each day includes:

- **2–3h theory** — deep explanations with diagrams
- **1–3h hands-on coding** — runnable Python examples
- **A concrete deliverable** — something you built and can demonstrate

🗓️ Syllabus Overview

Day	Topic	Key Concepts	Deliverable
Day 1	Modern LLM Foundations	Transformers, tokenization, RLHF, inference	Tokenization explorer + sampling visualizer
↳ Part A	Tokens & Embeddings	BPE, subword tokenization, embedding space	

↳ Part B	Transformer & Attention	Q/K/V, multi-head attention, FFN, residuals	
↳ Part C	Training & Inference	Positional encoding, SFT, RLHF/DPO, sampling	
↳ Appendix	Advanced Techniques	MoE, GQA, LoRA, quantization, scaling laws	
Day 2	Advanced Prompt Engineering	CoT, structured output, function calling	Reusable prompt library with validators
Day 3	Single-Agent Systems (ReAct)	ReAct loop, tools, memory types	Functioning ReAct agent with tools
Day 4	Planner/Executor Architecture	Task decomposition, critic loops, guardrails	Planner/Executor system with Critic
Day 5	RAG + Vector Memory	Embeddings, chunking, hybrid search	Agent with RAG + dynamic memory
Day 6	Multi-Agent Collaboration	Orchestration, agent roles, deadlocks	3-agent system solving real tasks
Day 7	Final Project: Build Your Copilot	Full system integration	Production-quality agentic system
Appendix	Real-World Agents: Cline & Roo Code	MCP, modes, memory, cost routing	Mental model of production agents

🧭 How to Use This Guide

1. **Follow in order** — each day builds on the previous
2. **Type every code example** — don't just read, build muscle memory
3. **Complete the deliverable** before moving to the next day
4. **Experiment** — modify the examples, break things, understand why

🗺️ Learning Path

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Day 1: LLM Foundations
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Day 2: Prompt Engineering
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Day 3: Single Agent (ReAct)     ← First agent!
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Day 4: Planner/Executor         ← Multi-component agent
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Day 5: RAG + Memory             ← Knowledge-grounded agent
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Day 6: Multi-Agent              ← Agent collaboration
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    ▼
Day 7: Final Project            ← Senior-level system
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Appendix: Cline & Roo Code     ← Production case studies
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Let's begin → Day 1: Modern LLM Foundations

Or jump to → Appendix: Real-World Agents (Cline & Roo Code)